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RESEARCH ARTICLE

Movement Disorder Evaluation of Parkinson's Disease Severity Based on Deep Neural Network Models

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ABSTRACT Parkinson's disease has been extensively studied using artificial intelligence for severity assessment. Previous research has achieved excellent results using wearable solutions. However, this approach has significant cost disadvantages, especially in the post-COVID-19 era. Recently, researchers are investigating vision-based alternatives for severity assessment. Unfortunately, the outcomes of these attempts often fall short when compared to wearable solutions. In this paper, we propose a novel feature extraction method suitable for temporal models, called 'Multitemporal Feature Average Pooling.' This method effectively captures crucial features from temporal data. Additionally, we employ interpretable artificial intelligence techniques to analyze the models, demonstrating their importance in severity assessment. We trained four severity classification models based on MDS-UPDRS. In the multi-class task of finger tapping, we achieved accuracies of 77% for the left hand and 85% for the right hand. These results represent an improvement of 4.6% and 12.6% compared to Shao et al.'s Three-Stream CNN model. In the binary classification, both left- and right-hand tasks achieved 92% accuracy, showing an increase of 11.4% and 14% compared to Chang et al. Furthermore, we achieved accuracies ranging from 60% to 80% for multi-class tasks involving Hand Movements, Rapid Alternating Movements of Hands, and Leg Agility. In binary classification, these tasks reached an even higher accuracy of 80% to 90%. This research is crucial for advancing vision-based severity assessment in clinical settings, providing more accurate evaluations for Parkinson's disease patients. Through the application of artificial intelligence technology, we anticipate achieving substantial breakthroughs and advancements in clinical practice.

INDEX TERMS Deep learning, LSTM, MDS-UPDRS, Parkinson's disease, severity classification.

I. INTRODUCTION

Parkinson's disease, the second most prevalent neurodegenerative disorder in elderly Americans [1], primarily affects the motor nervous system, showcasing symptoms like tremors, limb rigidity, reduced motor function, and abnormal gait [2].

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However, the current diagnosis relies on subjective clinical assessments, lacking objective tools [3]. The need for an objective assessment system is gaining attention.

In the post-COVID-19 era, many people avoid going to the hospital for treatment, the emergence of remote healthcare technology could solve this problem. This has led to increased research in remote health monitoring for Parkinson's disease [4], [5], [6].

Researchers have recently shifted their focus towards visual imagery due to the high cost associated with wearable solutions [6], [7], [8]. This method employs computer vision and AI to analyze patients' movements and postures. The advantage lies in using regular cameras, reducing the need for specialized sensors, making it more convenient and cost-effective. However, limitations persist, such as camera cost and the required photography skills. Our goal is to develop a smartphone-based visual imagery method using OpenPose [9] to extract joint point features from videos.

In [6] and [7], joint node coordinates, velocities, and distances were extensively calculated, aiding Parkinson's diagnosis. Similar to humans distinguishing cats and dogs, defining these differences explicitly is challenging. Our brains differentiate based on subtle details like shape, color, and contours. Similarly, we propose a novel approach using temporal models for feature extraction.

Our goal is to use counting models to extract features for the severity classification model. These features can offer elusive traits for the severity model's learning. Also, previous studies [6], [7] lacked distinguishing features for various severity levels, missing patients with specific symptoms. To tackle this, we will provide the severity model with features spanning a longer period, testing the feasibility through experiments.

In our paper, specialized medical professionals emphasized that movement count frequency variation is crucial for distinguishing severity levels. Consequently, we trained a counting model capable of determining the number of movements per ten seconds for each subject. During the training process, we ensured that the model learned from segments of the videos corresponding to the beginning, middle, and end portions, thereby avoiding overfitting due to excessively similar data. Next, we employed a sliding window approach to input the coordinate features for each ten-second interval into the counting model. The counting model then produced results, enabling the severity model to obtain the temporal changes in movement counts. Simultaneously, we extracted features from specific layers of the counting model to serve as input features for the severity model. These features are likely to contribute to severity assessment but are difficult for humans to define. To optimize the counting model's performance, we employed various techniques. By leveraging the outputted count features and utilizing the feature extraction method proposed in this study, we provided the severity model with robust features, aiming to improve prediction accuracy by supplying more abundant and powerful features.

Due to our observation that OpenPose predicts results that are highly unstable and imprecise at certain distances or angles, we have established filming standards for recording videos of Item 23 Finger Taps, Item 24 Hand Movements, Item 25 Rapid Alternating Movements of Hands, and Item 26 Leg Agility, as shown in Table 1. Through extensive testing, we have discovered that following the standards outlined in Table 1 leads to more accurate and stable predictions by OpenPose.

TABLE 1. Filming standards.

Item	Lens Height (cm)	Distance from Lens to Knee of subject (cm)	Filming angle
Finger Taps	110	100	head up
Hand Movements	110	100	head up
RAMOH ¹	210	120	overlook
Leg Agility	45	180	head up

¹ Rapid Alternating Movements of Hands

The Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS) [10], [11] was proposed by the International Parkinson and Movement Disorder Society in 2008 as a revised version of the UPDRS developed in the 1980s. The MDS-UPDRS is used to assess various aspects of Parkinson's disease. Because the MDS-UPDRS is widely applied as a standard in assessing Parkinson's disease, it is also frequently used as a benchmark in the field of artificial intelligence for Parkinson's disease assessment [6], [7]. In this paper, we primarily used the version revised by the Taiwan Movement Disorder Society in 2020 [12].

The following are the main contributions of this paper:

- 1) A new labeled image dataset for Movement Disorder Evaluation of Parkinson's Disease was collected for this work in collaboration with the doctors of the Neurology department at the Ditmanson Medical Foundation Chia-Yi Christian Hospital. The collection methods were for different evaluation actions and were standardized across all subjects. All images were labeled by the hospital practicing doctors with expertise in Parkinson's Disease and related Neurology. This new dataset consists of 253 video clips and when cropped into left and right (hand/leg), there are 506 video clips and labels. There were 177 videos of non-patient subjects (Level 0 in MDS-UPDRS classification) and the rest 329 videos were of patient subjects, which accounts for 65% of the total number of video samples. The dataset will be promoted a benchmark of research value to future work in this area.
- 2) All videos were pre-processed with opencv to ensure that the same number of JSON files could be generated per second, and for different test scenarios, we select appropriate features from the coordinates obtained after applying OpenPose to the videos.
- 3) A novel feature extraction method called Multitemporal Feature Average Pooling was proposed for temporal data based on the classic global average pooling. The core idea is to extract meaningful features from each time interval that are relevant to the model's objective and then perform summation and averaging to obtain information from all time intervals. From our experiments using Shapley additive explanations, it is shown that this method plays one of the most important roles in movement disorder severity level classification for all the evaluated actions.

- 4) Action-optimized models were proposed for action counting using Bi-Directional LSTM blocks. Each variant of the model was optimized for the specific action. Four different evaluation actions that were considered in this work and considering the left and right hand/leg actions, there were 8 counting models proposed. The counting models only differed in the number of neurons of different layers, but they had the same time window of 10 seconds video, that is 30 frames each (3 frames per second).
- 5) The results of this work will eventually be deployed and used by not only the professional doctors, but also caretakers of Parkinson's Disease for self-evaluation and personal guidance.

The remaining parts of this paper are organized as follows. In Section II, we introduce the application of artificial intelligence in Parkinson's disease. In Section III, we introduced the counting model, severity classification model, and a feature extraction method that is better suited for temporal models, as proposed by us. The experimental results of our proposed models are presented in Section IV, and the conclusion and future work are given in Section V.

II. RELATED WORKS

A. EVOLUTION OF ARTIFICIAL INTELLIGENCE APPLICATIONS IN PARKINSON'S DISEASE

In the past decade, various applications of artificial intelligence in the field of Parkinson's disease have emerged, and wearable solutions [8], [9], [10], [11] displaying precise dynamic signals have shown promising performance. Real-time objective monitoring and evaluation of the Bradykinesia and rigidity in Parkinson's disease enable physicians to diagnose and treat the disease with more objective and multiple evaluation criteria. For example, in [13], features were extracted from nine degrees of freedom sensors degrees of freedom sensors, and support vector machine techniques were used to classify finger tapping, diadochokinesis, and toe tapping items based on the MDS-UPDRS criteria with classification errors of approximately 10% to 20%. In [14], the authors collected gait data from Parkinson's disease patients and healthy individuals through a single-IMU foot-mounted wearable device and achieved excellent results in distinguishing whether a subject has Parkinson's disease with an F1 score of over 95% using support vector machine techniques. In [15], features were extracted from a smartwatch wearable device, including an accelerometer and a gyroscope. Nineteen features were extracted from the device, and decision tree, support vector machine, discriminant analysis, random forest, and k nearest neighbor techniques were used in the classification task of Tremor Severity. In [16] previously mentioned, good results were obtained using long short-term memory techniques. In our study, we will combine smartphones to try to improve the limitations of wearable devices required in [16].

Recently, there has been a growing shift in Parkinson's disease assessment towards vision-based methods [6], [7], [8]. These approaches have the advantages of reducing diagnostic costs and providing rich motion information, such as gait patterns, movement coordination, and body postures. Such information can be used to evaluate the severity of the disease and monitor its progression. For instance, in [8], Convolutional Pose Machines [17] were employed to extract key points from leg agility and toe-tapping videos, and random forest techniques were utilized to achieve an average F1-score of 0.631 and 0.725 for the two tasks, respectively. The results indicated that training models for Parkinson's disease tasks can be challenging, and factors like camera angles and zoom might influence the outcome. The authors suggested that adhering to standard specifications to eliminate camera shake could improve results.

Furthermore, they claimed that the limitations of Convolutional Pose Machines, which impacted video resolution, also affected their results. Thus, in our experiment, we adopted OpenPose for joint point extraction. Similarly, in [6], three cameras were placed at different positions to conduct several MDS-UPDRS assessments, including Finger Taps. Support vector machines and deep neural networks were employed for the binary classification of normal and abnormal cases, achieving an accuracy of 70% to 80%. This indicated that training models for such tasks can be difficult and may lack some crucial techniques or features. In [7], OpenPose was also used to obtain joint coordinates, and features such as coordinates, velocities, and Euclidean distances between points within a five-second window of Finger Taps videos were calculated. They established a three-stream fine-grained classification network with a Markov chain fusion model to aggregate multi-stream information from the skeleton sequence. Additionally, they proposed a Mini-batch-based balanced algorithm to address the data imbalance issue. Under complex model training, they achieved 60% to 70% accuracy in multi-class classification and introduced a new and more fault-tolerant metric called AcceptableACC, where accuracy reached 90%. Despite these achievements, there is still room for improvement compared to wearable solutions. In these studies, data scarcity, data imbalance, or other practical medical issues may pose difficulties in the vision-based Parkinson's disease assessment. For instance, [7] mentioned that data imbalance makes the task challenging, and [15] also pointed out that the uneven distribution of severity levels might affect classifier training and performance. Moreover, the low consistency in severity scores provided by two neurologists with respect to levels 3 and 4 further highlights the challenges in this domain. To address these limitations, we propose optimizing feature extraction. Our observations suggest that these studies may not have provided sufficient features to enable the models to distinguish severity levels effectively.

In recent years, there have been studies attempting to apply AI-assisted smartphone applications to Parkinson's disease. Due to the numerous attractive features of smartphones, these

applications resemble more like an extension of the device, rather requiring patients to purchase new equipment. In [18], various novel tests were custom-designed, such as touch screen tests, to record data for Parkinson's disease assessment using artificial intelligence. In [19], the inertial sensors embedded in smartphones were utilized for gait assessment in Parkinson's disease patients. The ubiquitous nature of smartphones can provide clinicians with objective long-term measurement results of the severity of symptoms, as well as record patients' symptom fluctuations and changes outside of the clinic. Moreover, it can allow patients to undergo assessments at any location according to their needs. However, these studies are based on different and newly defined measurement methods. Therefore, we attempt to leverage the convenience and advantages brought by smartphones and apply them to the classic MDS-UPDRS assessment.

B. OPENPOSE

Pose estimation is an important technology that can be applied in fields such as human-computer interaction, motion analysis, and surveillance. OpenPose is an open-source pose estimation library that uses deep learning techniques to achieve real-time estimation of human body poses. OpenPose, developed by researchers at the University of California, Los Angeles, employs a deep neural network model to extract human body pose information from a single image. One of the main advantages of OpenPose is that it can achieve pose estimation in different environments, such as indoor and outdoor environments, under different lighting conditions, and from different viewpoints. OpenPose mainly utilizes convolutional neural network techniques, as well as a technique called Part Affinity Fields, which can capture the relationships between different body parts to better estimate human body poses.

Parkinson's disease is a neurodegenerative disease that causes movement disorders such as limb stiffness and tremors. Therefore, posture detection is of great importance for the diagnosis and treatment of Parkinson's disease. We will use the joint coordinates provided by OpenPose as important features for our motion count and severity models.

C. SHAPLEY ADDITIVE EXPLANATIONS VALUES

Shapley additive explanations values [20] are a widely used technique for explaining the predictions of artificial intelligence models. Shapley additive explanations values provide a way to attribute a model's prediction to its input features by calculating the contribution of each feature to the model's output. Shapley additive explanations values have several advantages over other feature attribution methods, including being less constrained by the model, providing both global and local interpretability, and being able to handle interactions between features. Moreover, shapley additive explanations values can be used to generate various feature relationship plots, which provide significant help for the interpretability of the model.

D. SUMMARY

This section provides an overview of the evolution of artificial intelligence applications in Parkinson's disease assessment and discusses the differences and outcomes among various methods. Table 2 summarizes recent research on wearable solutions for Parkinson's disease assessment, while Table 3 summarizes research on vision-based solutions. These tables present the assessment tasks, models, techniques, features, and outcomes of the respective studies.

III. METHODS

We will provide a detailed explanation of the overall process and methods we employed to assess Parkinson's disease, as depicted in Fig. 1.

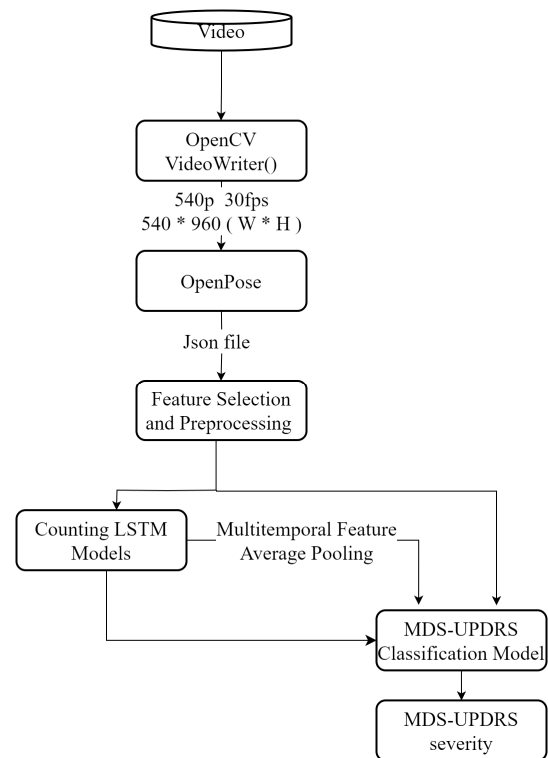


FIGURE 1. Overall architecture of deep learning models based movement disorder evaluation of parkinson disease severity.

A. FEATURE SELECTION AND PREPROCESSING

Before feature selection and preprocessing, we preprocess all the videos using OpenCV [21] to convert them into 540p videos with a frame rate of 30 and an aspect ratio of 540:960. This ensures that all videos, after undergoing OpenPose, generate an equal number of JavaScript object notation files per second. Furthermore, we normalize the coordinate values within a consistent range.

For different test scenarios, we select appropriate features from the coordinates obtained after applying OpenPose to the videos. These features serve as input for the LSTM Counting Model and MDS-UPDRS Classification Model, as shown in Table 4.

TABLE 2. Summary of wearable solutions for movement disorder evaluation in parkinson disease.

Research	Year	Evaluation Item	Model	Characteristics	Results
Manzanera et al. [13]	2015	Finger Tapping	SVM	Orientation sensor	classification errors:15–16.5%
Manzanera et al. [13]	2015	Toe Tapping	SVM	Orientation sensor	classification errors:18.2–20.2%
Manzanera et al. [13]	2015	Diadochokinesis	SVM	Orientation sensor	classification errors:9.3–9.8%
Jeon et al. [15]	2017	Tremor	Decision tree	Wrist-watch-type wearable device UPDRS	±85.55%
Pérez-Ibarra et al. [14]	2020	Gait	SVM	single-IMU foot-mounted wearable device Binary classification:Healthy/PD	F1-score: Healthy :0.987 PD :0.953%
Balaji et al. [16]	2021	Gait	LSTM	LSTM VGRF Sensor H&Y scale Binary classification:Healthy&PD	Accuracy: 98.6%

TABLE 3. Summary of vision-based solutions for movement disorder evaluation in parkinson disease.

Research	Year	Evaluation Item	Model	Characteristics	Results
Li et al. [8]	2018	Leg Agility	Random Forest	Convolutional Pose Machines	F1-score: Right:0.59 ± 0.01 Left :0.73 ± 0.04
Chang et al. [6]	2019	Finger Taps	SVM DNN	OpenPose UPDRS Binary classification:Healthyabnormal (0) / (1-4)	Accuracy: Right :78.01% Left :80.60%
Shao et al. [7]	2021	Finger Taps	Three-Stream CNN (Markov chain spatial-temporal attention CNN FC) SVM	OpenPose MDS-UPDRS mini-batch-based balanced	Accuracy: SVM :61.6% Three-Stream CNN :72.4%

TABLE 4. Features input for each task.

Item	left hand keypoints 2d	right hand keypoints 2d	pose keypoints 2d
FT ¹ Right		1-21	
FT Left	1-21		
HM ² Right		1-21	
HM Left	1-21		
RAMOH ³ Right	1-21		3-4
RAMOH Left	1-21		6-7
LA ⁴ Right			9-11, 22-24
LA Left			12-14, 19-21

¹ Finger Taps ² Hand Movements ³ Rapid Alternating Movements of Hands ⁴ Leg Agility

During the feature selection and preprocessing stages, it mainly involves coordinate initialization and data standardization. In terms of data standardization, we apply z-score normalization to each feature. The formula is shown as follows:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Here z represents the transformed data value, x represents the original data value, μ represents the average number of each joint point selected according to the task, and σ represents the standard deviation value of each joint point selected according to the task.

The purpose of coordinate initialization is to ensure that when subjects perform the same action at different positions in the video, the resulting coordinate values do not differ

significantly. Therefore, we primarily select positions that are less likely to move during the action tests as reference points, using the initial frame as a baseline.

For the Finger Taps and Hand Movements tests, we select the 0th point of the left and right hands as the reference points. In the Rapid Alternating Movements of Hands test, we choose the first point of the body as the reference point. In the Leg Agility test, we use the 8th point of the body as the reference point. Through data observation, we have found that these joint positions are relatively fixed and stable across the different tasks.

B. LSTM COUNTING MODEL

According to our co-authors who are practicing doctors in the hospital, the judgment of Parkinson's disease is often based on the action stability and speed during the movement, which means that it is a time series problem. LSTM models are known to perform quite well in learning the temporal characteristics from time-series data. We tried with LSTM and then we customized it into a bi-directional version which performed quite well for our dataset. During testing, the RMSE error rates for LSTM ranged around from 0.683 to 5.71, which is quite insignificant and does not affect our final results, thus LSTM models were chosen. As per the concept mentioned in [22], we should guarantee that the network consistently generates activations with the desired distribution, regardless of the parameter values. Therefore, in the design of the LSTM Counting Model, we will use bidirectional long short-term memory, followed

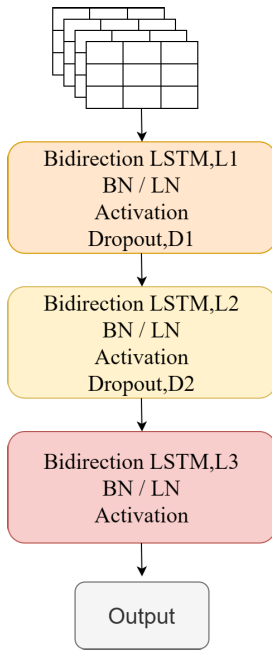


FIGURE 2. Architecture of counting LSTM models.

by normalization, activation, and dropout, in that order, as depicted in Fig. 2 and Algorithm 1.

Since our goal is to improve the severity model by extracting excellent features, it is crucial to optimize the counting model. Thus, during the normalization step, we will explore the use of batch normalization (BN) and layer normalization (LN). For the activation function, we will experiment with options such as tanh, rectified linear unit, leaky rectified linear unit, parametric rectified linear unit, and exponential linear unit. We will then select the best-performing model from these options.

For each task, we have designed appropriate numbers of neurons and dropout probabilities for each layer based on Fig. 2, as shown in Table 5. These configurations have been determined through a combination of rationality and iterative refinement.

TABLE 5. Numbers of neurons and dropout probabilities for each layer of the counting model.

Item	L1	D1	L2	D2	L3
Finger Taps Right	30	0.5	15	0.3	5
Finger Taps Left	30	0.4	15	0.2	5
Hand Movements Right	30	0.6	15	0.3	5
Hand Movements Left	30	0.6	15	0.3	5
Rapid Alternating Movements of Hands Right	30	0.5	15	0.3	5
Rapid Alternating Movements of Hands Left	30	0.5	15	0.3	5
Leg Agility Right	10	0.5	5	0.3	2
Leg Agility Left	10	0.5	5	0.3	2

According to professional advice, understanding the variation of action frequency over time is crucial in the medical field. We believe that a great action counting model can provide excellent input features to the severity model.

To achieve this, we adopted the least common multiple of five seconds and ten seconds, which is ten seconds, as the video input length for the counting model. The purpose of this choice is to avoid having durations that are too short, which may not provide sufficient features, and also to prevent durations that are too long, which could lead to significant differences in counting labels that are difficult to learn from.

Moreover, we allowed the counting model to learn from various ten-second segments of the videos, such as from zero seconds to ten seconds and from ten seconds to twenty seconds. Additionally, through data observation, we found that in more severe cases, patients tend to stop recording videos after experiencing severe shaking or when there is no active movement. Consequently, we consider the content of the last few seconds of the video to be more important. To ensure that the model can effectively learn these crucial segments, if the content of the last ten seconds overlaps with the middle ten seconds by more than five seconds, we choose to remove the middle ten seconds. For example, if the total video length is twenty-six seconds, the model will learn features from the segments zero seconds to ten seconds, ten seconds to twenty seconds, and sixteen seconds to twenty-six seconds. If the video is only twenty-four seconds long, the model will learn features from the segments zero seconds to ten seconds and fourteen seconds to twenty-four seconds.

The purpose of this approach is to avoid having similar features that exceed five seconds in the dataset, which could cause overfitting of the model. At the same time, it allows the model to possess sufficient predictive capability for the entire video. During this learning process, we did not utilize a sliding window data partitioning method, such as from zero seconds to ten seconds or one second to eleven seconds. This decision is made to prevent the model from learning excessively similar content, thus avoiding overfitting on the dataset.

C. MULTITEMPORAL FEATURE AVERAGE POOLING

In this paper, we propose a novel feature extraction method based on a temporal model called multitemporal feature average pooling, as illustrated in Fig. 3. Our inspiration comes from classic global average pooling, and we attempt to apply this approach to temporal models. The core idea of our proposed multitemporal feature average pooling is to extract all neurons from different time intervals of the temporal model. This means that we extract meaningful features from each time interval that are relevant to the model's objective and then perform summation and averaging to obtain information from all time intervals. After feeding the coordinate information into the counting model, we calculate the average of the activation values from the three LSTM layers of the counting model according to the corresponding time segments to calculate the average value of the corresponding activation for each segment.

Ensuring the alignment of the LSTM counting model, which utilizes multitemporal feature average pooling for feature extraction, and the MDS-UPDRS classification model,

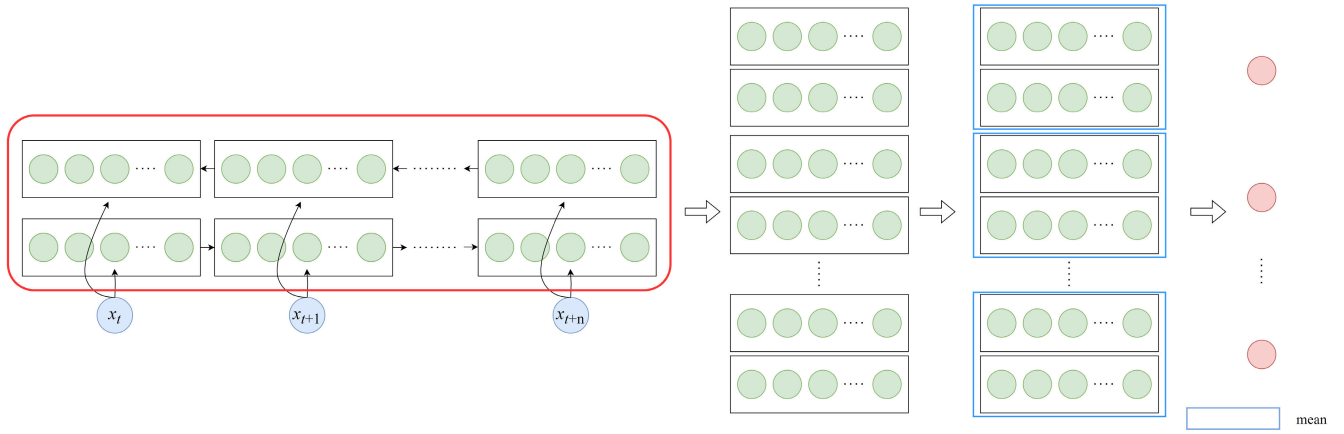


FIGURE 3. Conceptual image of multitemporal feature average pooling.

Algorithm 1 LSTM Counting Model

Require: Adopt different joint combinations according to different actions. In units of 10 seconds.

Ensure: leg_times_predict_for_mds_updrs

- 1: Pre-processing of coordinates into LSTM inputs.
- 2: Initialize Sequential Model.
- 3: Add First Bidirectional LSTM Layer {
Add a Bidirectional LSTM layer (neurons' number = 10)
Apply L2 regularization on kernel and recurrent weights.
(kernel_regularizer = l2(0.05), recurrent_regularizer = l2(0.01))
Set return_sequences=True.
}
- 4: Add Layer Normalization.
- 5: Add PReLU Activation.
- 6: Add Dropout Layer (Set the dropout rate to 0.5)
- 7: Add Second Bidirectional LSTM Layer {
Add a Bidirectional LSTM layer (neurons' number = 5)
Apply L2 regularization on kernel and recurrent weights.
(kernel_regularizer = l2(0.05), recurrent_regularizer = l2(0.01))
Set return_sequences=True.
}
- 8: Add Layer Normalization.
- 9: Add PReLU Activation.
- 10: Add Dropout Layer (Set the dropout rate to 0.3)
- 11: Add Third Bidirectional LSTM Layer {
Add a Bidirectional LSTM layer (neurons' number = 2)
Apply L2 regularization on kernel and recurrent weights.
(kernel_regularizer = l2(0.05), recurrent_regularizer = l2(0.01))
Set return_sequences=False.
}
- 12: Add Layer Normalization.
- 13: Add PReLU Activation.
- 14: Add Dense layer.
- 15: Print the model summary

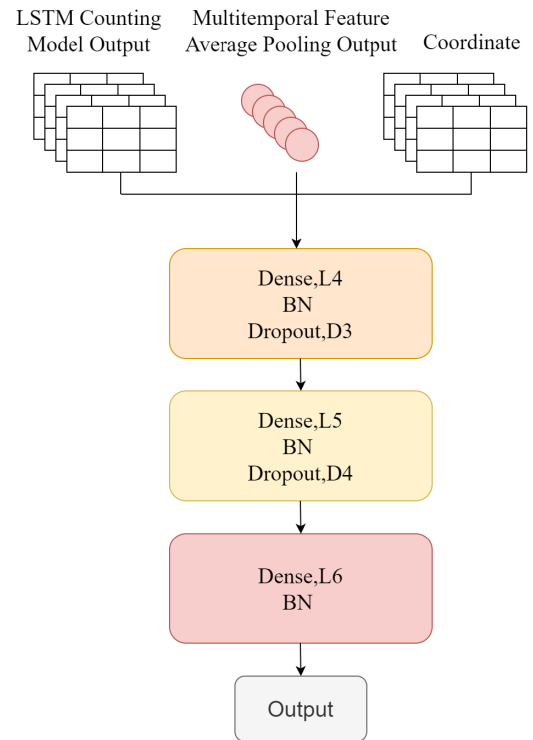


FIGURE 4. Architecture of MDS-UPDRS classification model.

which takes these features as input, is crucial as they serve the same task. This indicates a high degree of correlation between them. We can expect these extracted, relevant features to provide numerous undefined and meaningful characteristics for similar tasks. Analogous to our ability to distinguish between cat ears and dog ears without explicitly defining the differences, our method utilizes information from all time periods, granting these features the capability to incorporate insights from each time segment equally.

Furthermore, although Fig. 3 showcases a bidirectional model, in practice, this approach can also be applied to unidirectional temporal models.

D. MDS-UPDRS CLASSIFICATION MODEL

To validate our previous statement that previous studies in this domain did not provide enough and sufficiently good features, we aimed to keep the architecture of the MDS-UPDRS severity classification model as simple and intuitive as possible. We achieved this by using only a few layers of deep neural networks, as depicted in Fig. 4 and Algorithm 2. Our focus in the experiments was primarily on the input features. The model takes inputs from the LSTM Counting Model's outputs, features extracted by multitemporal feature average pooling, and basic coordinate information.

Algorithm 2 UPDRS Classification Model

Require: output from countingLSTM, output from multitemporal feature average pooling, coordinates of the joints

Ensure: UPDRS classification

- 1: Initialize Input layer.
 - 2: Add First Dense Layer {
Add a Dense layer with 64 neurons
}
 - 3: Add Batch Normalization.
 - 4: Add Dropout Layer {
Set the dropout rate to 0.5
}
 - 5: Add Second Dense Layer {
Add a Dense layer with 32 neurons
}
 - 6: Add Batch Normalization.
 - 7: Add Dropout Layer {
Set the dropout rate to 0.3
}
 - 8: Add Third Dense Layer {
Add a Dense layer with 16 neurons
}
 - 9: Add Batch Normalization.
 - 10: Print the model summary
-

The detailed architecture follows the following sequence: dense layer L4, batch normalization, dropout layer D3, dense layer L5, batch normalization, dropout layer D4, dense layer L6, batch normalization, and the classification layer. Based on Fig. 4, we have designed appropriate numbers of neurons and dropout probabilities for each layer, as shown in Table 11 (see Appendix). These configurations were determined based on rationality. As we aim for the results to be derived from the provided features rather than from iteratively adjusting parameters, we did not extensively fine-tune the hyperparameters at this stage.

In the MDS-UPDRS classification model, we observed that the cycle of each of the four actions is approximately one second. Therefore, we will process the coordinates for each ten-second segment with a sliding window on a per-second basis. Each ten-second segment's coordinates will be input into the LSTM counting model, and the output will be

the action frequency for each ten-second segment, which will serve as a feature input for the MDS-UPDRS classification model. Simultaneously, we will use our proposed multitemporal feature average pooling method to extract features from each bidirectional long short-term memory layer as another set of feature inputs for the MDS-UPDRS classification model. Additionally, we will include the most fundamental coordinate information as inputs.

The sliding window will continuously slide from 0 to 10 seconds to 10 to 20 seconds. This means that the MDS-UPDRS severity classification model will receive inputs of action frequencies for each ten-second segment within the 20-second window, such as 0 to 10 seconds and 1 to 11 seconds. Simultaneously, it will use the features extracted through our proposed multitemporal feature average pooling method and the selected coordinate feature values within the 20-second window based on the task requirements.

We chose to provide features from a 20-second video, which is twice as long as [6] and four times as long as [7]. The aim is to provide sufficient features for distinguishing severity by using longer video segments.

IV. RESULTS

A. EXPERIMENTS

1) EXPERIMENTAL ENVIRONMENT

A GPU server is used in this paper, Table 6 shows the server's hardware specifications and the programming languages used in this paper.

TABLE 6. Details of the GPU server specifications.

Hardware	Specifications
CPU	Intel (R) Xeon (R) Silver 4110 CPU @ 2.10GHz
Memory	128GB RAM
GPU	GeForce RTX 2080 Ti 11GB
Operating System	Ubuntu 16.04.6 LTS 64-bit
Programming Environment	Python 3.6.8, Tensorflow 1.13.1, Keras 2.3.0

2) DATASET

In this section, we will describe the datasets and input features used in the LSTM counting model and MDS-UPDRS classification model in the paper. The data used in our experiments were obtained from MDS-UPDRS test videos recorded at Chiayi Christian Hospital from 2021 to 2023. Additionally, we maintained an 8:2 ratio for training and testing sets in all stages of our study.

First, we will describe the counting-related information of the LSTM counting model dataset in Table 10 (see Appendix). Number of videos column represents the total number of videos for each task, while number of video clips denotes the total number of data obtained by splitting the videos into ten-second segments, as mentioned earlier. For instance, it includes data from 0 to 10 seconds, 10 seconds to 20 seconds, and the last 10 seconds. The remaining columns represent the maximum, minimum, average, and standard

deviation of the action frequencies within these ten-second data segments.

In Table 7, we describe the number of videos for each severity level in the MDS-UPDRS classification model dataset, as well as the total number of videos. Since the physicians only label the severity level for one randomly selected video per day of motion testing and we have removed videos shorter than twenty seconds, the number of videos in this stage will be smaller compared to the counting model dataset.

TABLE 7. Number of videos of each severity level and total number of videos in MDS-UPDRS classification model datasets.

Item	Lv ¹ 0	Lv 1	Lv 2	Lv 3	Lv 4	Number of videos
Finger Taps Right	24	15	24	1	0	64
Finger Taps Left	18	34	10	2	0	64
Hand Movements Right	24	33	8	0	0	65
Hand Movements Left	26	33	6	0	0	65
Rapid Alternating Movements of Hands Right	20	36	8	1	0	65
Rapid Alternating Movements of Hands Left	22	36	6	1	0	65
Leg Agility Right	21	28	8	2	0	59
Leg Agility Left	22	29	6	2	0	59

¹ Level

B. EXPERIMENTAL RESULTS OF LSTM COUNTING MODEL

As shown in Table 2, the experimental results of the LSTM counting model are presented, which include the standard of root-mean-square error (RMSE) and times. In the finger tapping task, which exhibits significant variations in the number of taps, the model achieved an average error of approximately four counts. Similarly, in the Hand Movements task, the model performed well with an average error of less than two counts. Furthermore, in the Leg Agility task and Rapid Alternating Movements of Hands task, the model achieved excellent results with an average error of less than one count.

C. EXPERIMENTAL RESULTS OF MDS-UPDRS CLASSIFICATION MODEL

As mentioned in [15], there is a certain degree of variability among doctors in interpreting the severity levels. We encountered similar situations in our research. Therefore, in addition to presenting the original multi-class classification of MDS-UPDRS severity, we further binaryize the scores of each assessment task. The scores are divided into two categories: [0-1] representing mild and [2-3] representing moderate to severe. Table 9 shows the results of both the multi-class and binary classifications for each task.

In the Finger Taps task, excellent accuracy was achieved for both the right hand (85%) and the left hand (77%), in the multi-class classification. Moreover, in the binary classification, the accuracy reached an impressive 92% for

TABLE 8. Results of the LSTM counting model.

Item	Testing			Training		
	RMSE	ME ¹	MaxE ²	RMSE	ME	MaxE
FT ³ Right	5.71	4.286	16	3.611	2.699	10
FT Left	5.121	4.119	12	2.033	1.38	8
HM ⁴ Right	2.33	1.795	6	1.49	0.914	9
HM Left	2.215	1.705	7	1.239	0.678	6
RAMOH ⁵ Right	1.163	0.818	3	0.827	0.393	5
RAMOH Left	1.201	0.909	3	0.515	0.168	3
LA ⁶ Right	0.683	0.474	1	0.535	0.3	3
LA Left	0.684	0.421	2	0.518	0.3	2

¹ Mean Error (in times) ² Maximum Error (in times) ³ Finger Taps

⁴ Hand Movements ⁵ Rapid Alternating Movements of Hands ⁶ Leg Agility

both hands. In the Hand Movements task, an accuracy of 77% was achieved for both the right and left hands in the multi-class classification, while in the binary classification, the accuracies were 85% and 92%, respectively.

In the Rapid Alternating Movements of Hands task, the best accuracy achieved for the right hand was 69%, and for the left hand, it was 62% in the multi-class classification. In the binary classification, both hands reached excellent accuracies of 92%.

Finally, in the Leg Agility task, the best accuracy obtained for the right leg was 58%, and for the left leg, it was 50% in the multi-class classification. In the binary classification, impressive accuracies of 92% and 83% were achieved for the right and left legs, respectively.

The multi-class and binary confusion matrices for the Finger Taps task are shown in Fig. 5. The results for the Hand Movements task, the Rapid Alternating Movements of Hands task, and the Leg Agility task are shown in Appendix.

D. SHAPLEY ADDITIVE EXPLANATIONS VALUES RESULT OF MDS-UPDRS CLASSIFICATION MODEL

In the MDS-UPDRS classification model, the feature representing the coordinates from the A-th position at time T seconds is denoted as T_A. Additionally, the input of the coordinates from T seconds to T+10 seconds, processed by the LSTM Counting Model, generates a feature representing the count within that time period, which is named as T-T+10 second. Furthermore, utilizing our proposed multitemporal feature average pooling method, features extracted from the

TABLE 9. Results of the MDS-UPDRS classification model.

Item	Accuracy	Binary Task Accuracy
Finger Taps Right	85%	92%
Finger Taps Left	77%	92%
Hand Movements Right	77%	85%
Hand Movements Left	77%	92%
Rapid Alternating Movements of Hands Right	69%	92%
Rapid Alternating Movements of Hands Left	62%	92%
Leg Agility Right	58%	92%
Leg Agility Left	50%	83%

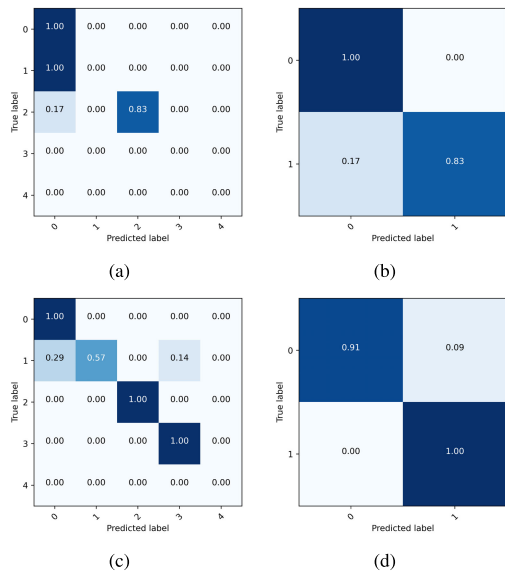


FIGURE 5. (a) Confusion matrix of right finger taps; (b) Binary confusion matrix of right finger taps; (c) Confusion matrix of left finger taps; (d) Binary confusion matrix of left finger taps.

L-th layer of the bi-directional long short-term memory in the LSTM Counting Model are named as lstmL_T-T+10s.

The severity levels and overall Shapley additive explanation values for the Finger Taps task in the right hand are shown in Fig. 6 (See Appendix for details.), while the results for the left hand are presented in Appendix. The contribution of features for the Hand Movements task, the Rapid Alternating Movements of Hands task, and the Leg Agility task, both in terms of severity levels and overall contribution, for the right and left sides can also be found in Appendix.

From the aggregated results across severity levels, it can be observed that the features extracted using our proposed multitemporal feature average pooling method often contribute significantly to the severity model. This is particularly evident

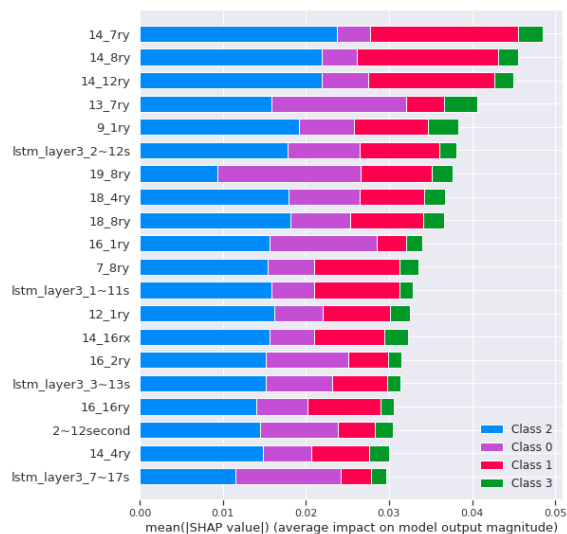


FIGURE 6. Shap value of each level of the right finger taps.

in the Finger Taps task, the right hand's Rapid Alternating Movements of Hands task, and the right leg's Leg Agility task.

Furthermore, for each severity level, these features are found to make excellent contributions. Additionally, it was discovered that the various features extracted from ten seconds onward also provide valuable information.

V. CONCLUSION

In this paper, we propose a novel feature extraction method called multitemporal feature average pooling, which is more suitable for temporal models. This method enables

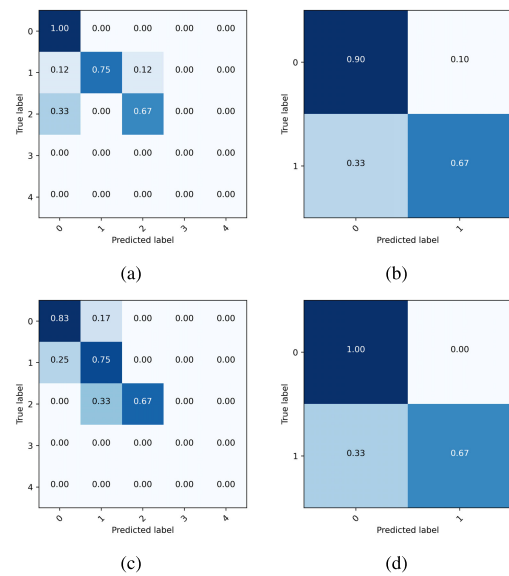


FIGURE 7. (a) Confusion matrix of right Hand Movements; (b) Binary confusion matrix of right Hand Movements; (c) Confusion matrix of left Hand Movements; (d) Binary confusion matrix of left Hand Movements.

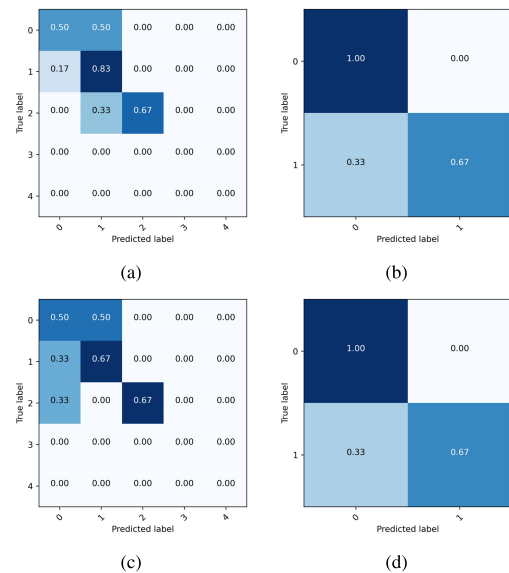


FIGURE 8. (a) Confusion matrix of right Rapid Alternating Movements of Hands; (b) Binary confusion matrix of right Rapid Alternating Movements of Hands; (c) Confusion matrix of left Rapid Alternating Movements of Hands; (d) Binary confusion matrix of left Rapid Alternating Movements of Hands.

TABLE 10. Details of LSTM counting model datasets.

Item	Maximum (times)	Minimum (times)	Mean (times)	Standard deviation (times)	Number of video clips	Number of video
Finger Taps Right	57	9	32.851	11.278	208	70
Finger Taps Left	51	8	31.716	10.802	208	70
Hand Movements Right	26	6	13.495	3.322	218	73
Hand Movements Left	27	6	13.509	3.347	218	73
Rapid Alternating Movements of Hands Right	15	3	8.7	2.428	217	75
Rapid Alternating Movements of Hands Left	15	3	8.7	2.436	217	75
Leg Agility Right	9	2	5.319	1.196	188	62
Leg Agility Left	9	1	5.335	1.189	188	62

TABLE 11. Numbers of neurons and dropout probabilities for each layer of the MDS-UPDRS classification model.

Item	L4	D3	L5	D4	L6
Finger Taps Right	256	0.5	64	0.3	16
Finger Taps Left	256	0.5	64	0.3	16
Hand Movements Right	256	0.5	64	0.3	16
Hand Movements Left	256	0.5	64	0.3	16
Rapid Alternating Movements of Hands Right	256	0.5	64	0.3	16
Rapid Alternating Movements of Hands Left	256	0.5	64	0.3	16
Leg Agility Right	64	0.5	32	0.3	16
Leg Agility Left	64	0.5	32	0.3	16

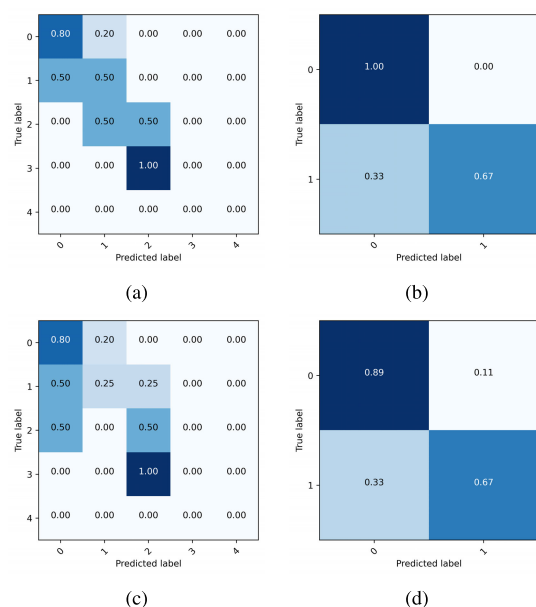
us to fairly extract feature information from each time segment, allowing the features to incorporate global temporal dependencies. It can be applied to both unidirectional and bidirectional temporal models. Additionally, we validate the effectiveness of the extracted features using Shapley additive explanation values. These features provide powerful but difficult-to-define features for severity classification tasks, leading to excellent performance. Moreover, this concept and method can be applied to other time-varying related tasks, eliminating the need for complex feature definition standards. Our study holds potential in advancing artificial intelligence applications for Parkinson's disease, promising more objective and accurate clinical assessments. In the post-COVID-19 era, this gains significant relevance, facilitating remote healthcare services and enhancing the objectivity and precision of clinical evaluations.

APPENDIX EXPERIMENTAL DETAILS

A. DATASETS AND ARCHITECTURES

The details of LSTM counting model datasets are shown in Table 10.

The numbers of neurons and dropout probabilities for each layer of the MDS-UPDRS classification model are shown in Table 11.

**FIGURE 9.** (a) Confusion matrix of right Leg Agility; (b) Binary confusion matrix of right Leg Agility; (c) Confusion matrix of left Leg Agility; (d) Binary confusion matrix of left Leg Agility.

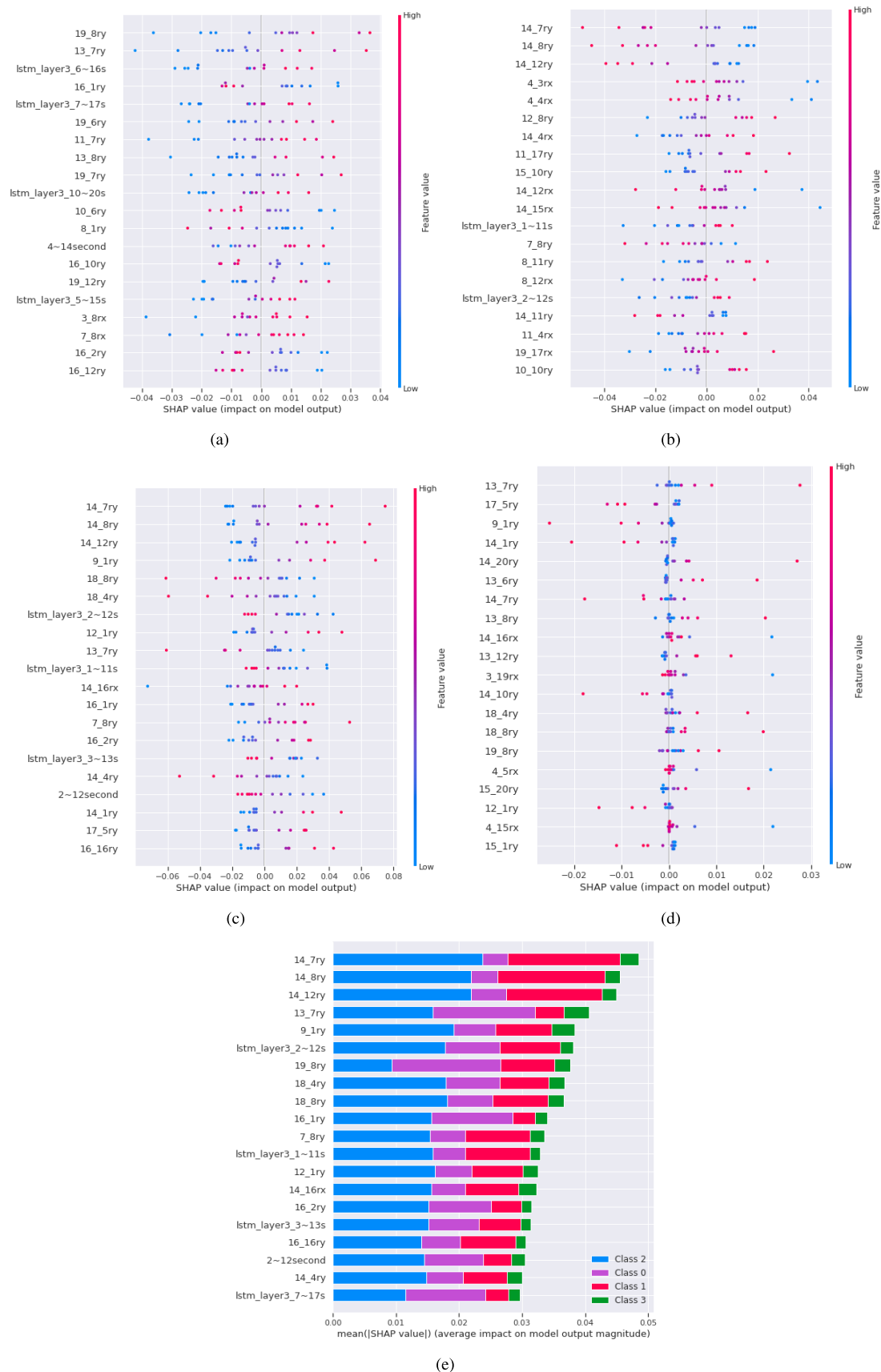


FIGURE 10. (a) Level 0 shap value of the right Finger Taps; (b) Level 1 shap value of the right Finger Taps; (c) Level 2 shap value of the right Finger Taps; (d) Level 3 shap value of the right Finger Taps. (e) Shap value of each level of the right Finger Taps.

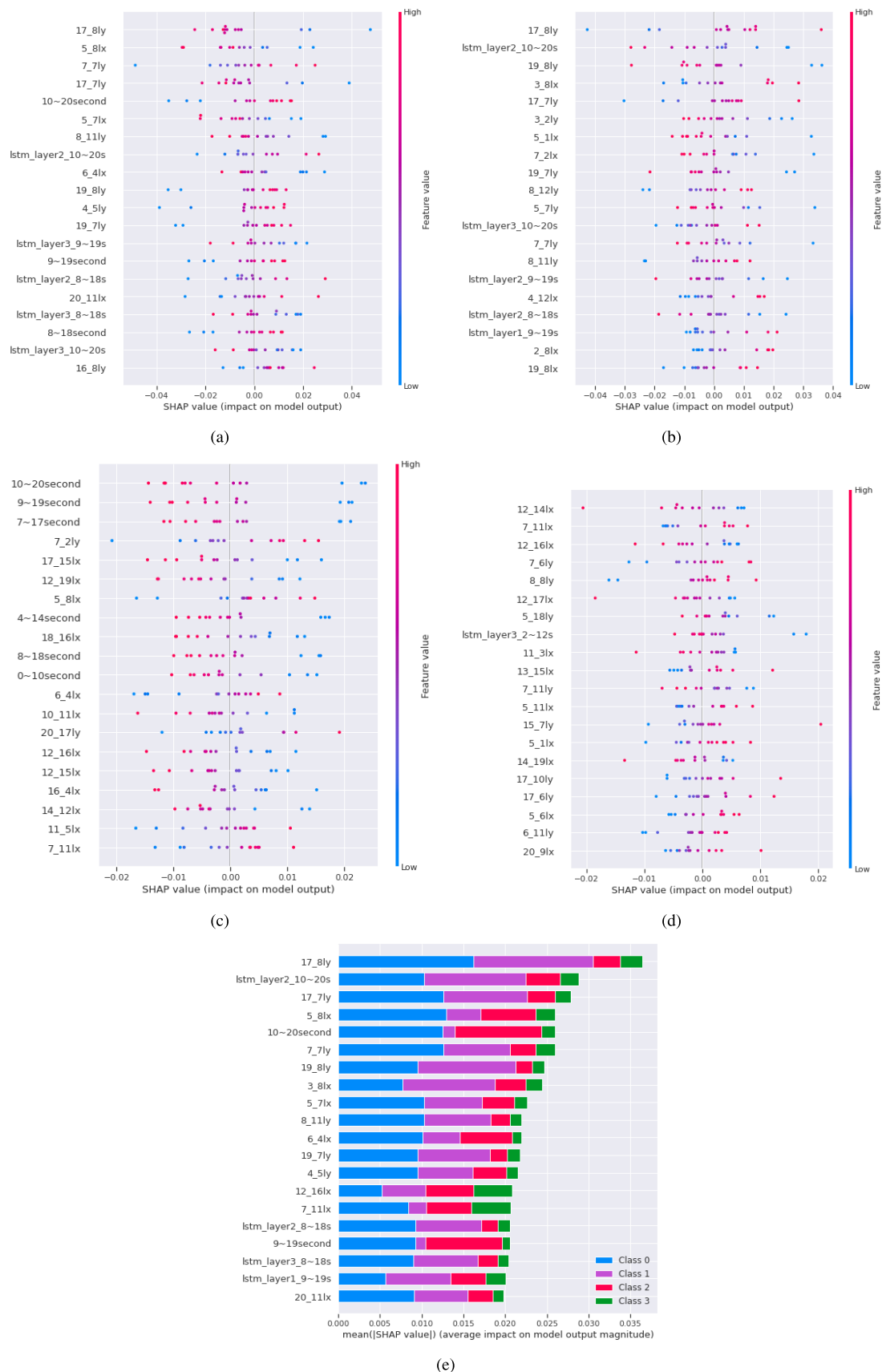


FIGURE 11. (a) Level 0 shap value of the left Finger Taps; (b) Level 1 shap value of the left Finger Taps; (c) Level 2 shap value of the left Finger Taps; (d) Level 3 shap value of the left Finger Taps. (e) Shap value of each level of the left Finger Taps.

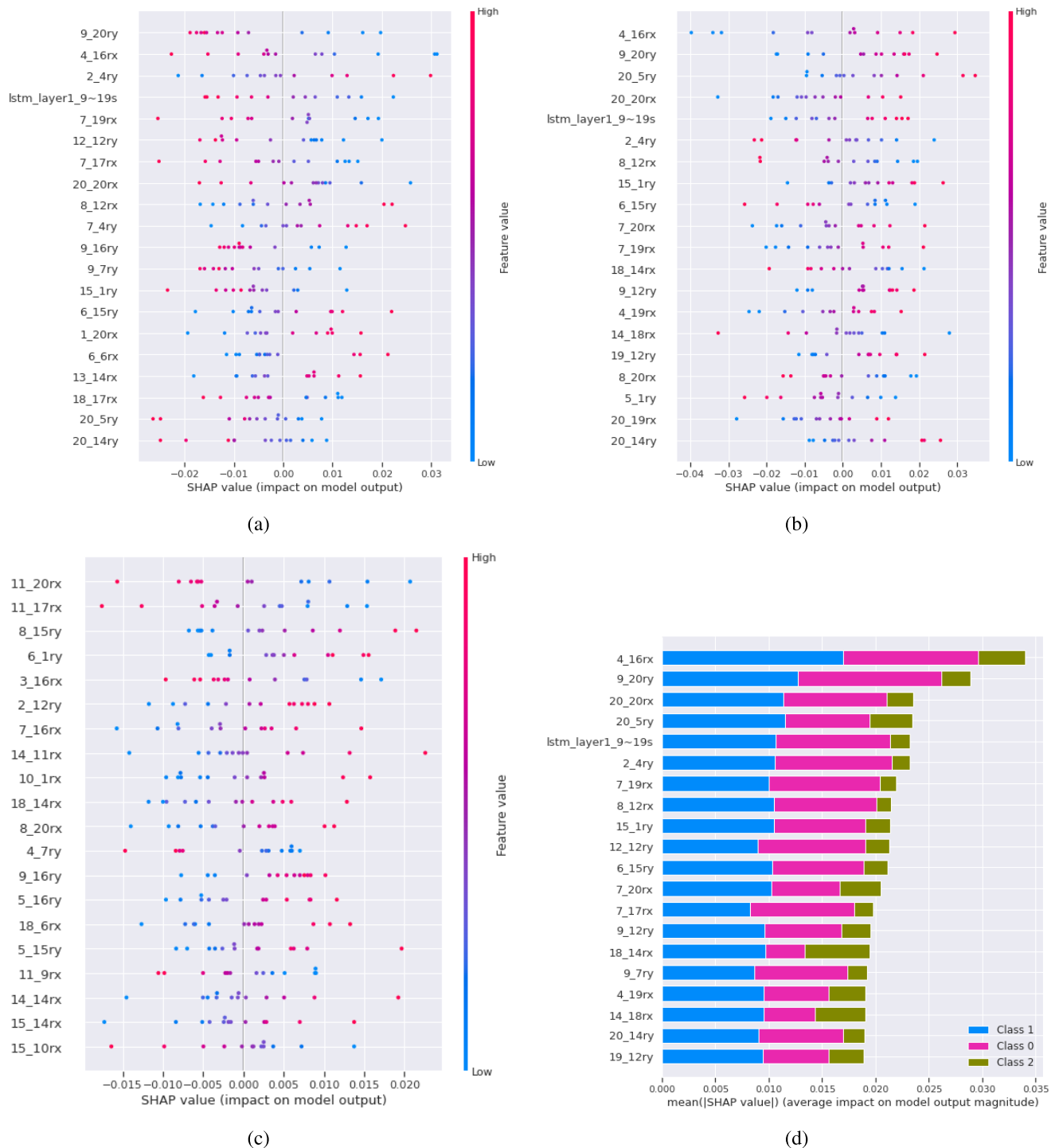


FIGURE 12. (a) Level 0 shap value of the right Hand Movements; (b) Level 1 shap value of the right Hand Movements; (c) Level 2 shap value of the right Hand Movements; (d) Shap value of each level of the right Hand Movements.

B. CONFUSION MATRICES

The multi-class and binary confusion matrices for the Hand Movements task are shown in Fig. 7. Fig. 8 display the confusion matrix results for the Rapid Alternating Movements of Hands task. Lastly, the confusion matrix results for the Leg Agility task are presented in Fig. 9.

C. SEVERITY LEVELS AND OVERALL SHAPLEY ADDITIVE EXPLANATION

Figs. 10–17 show the severity levels and overall Shapley additive explanation values for the Finger Taps, Hand Movements, Rapid Alternating Movements of Hands, and Leg Agility task.

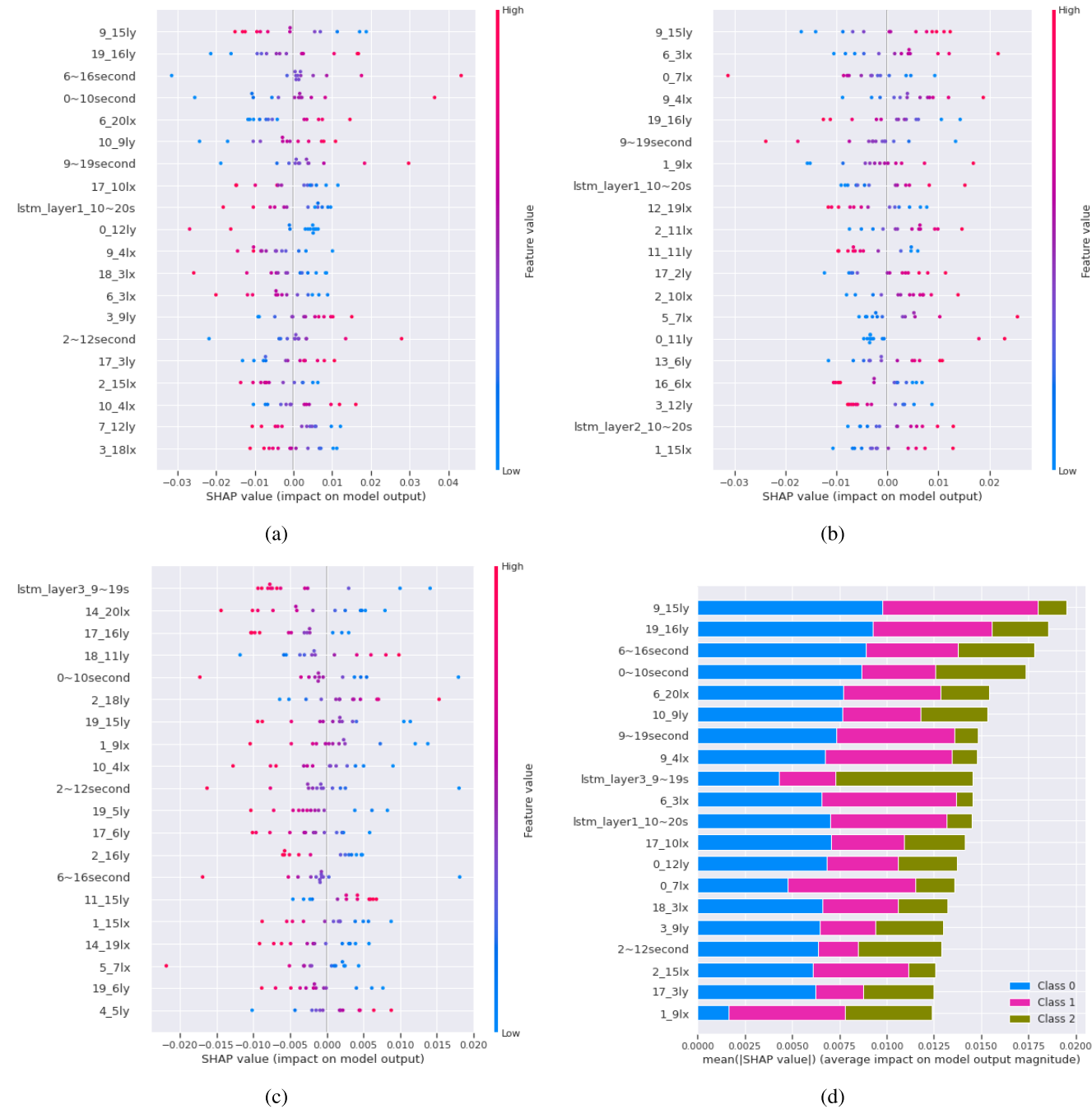


FIGURE 13. (a) Level 0 shap value of the left Hand Movements; (b) Level 1 shap value of the left Hand Movements; (c) Level 2 shap value of the left Hand Movements; (d) Shap value of each level of the left Hand Movements.

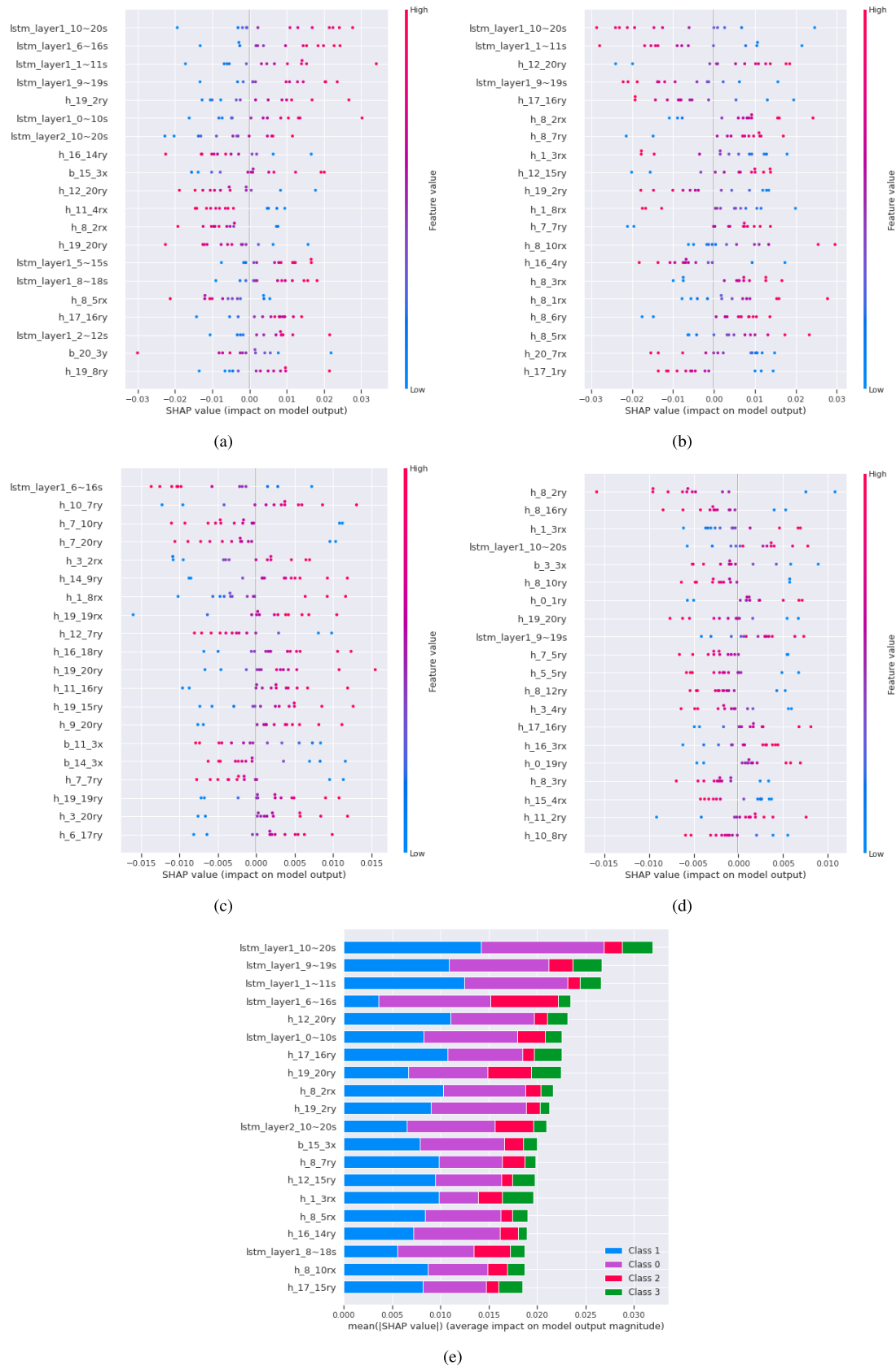


FIGURE 14. (a) Level 0 shap value of the right Rapid Alternating Movements of Hands; (b) Level 1 shap value of the right Rapid Alternating Movements of Hands; (c) Level 2 shap value of the right Rapid Alternating Movements of Hands; (d) Level 3 shap value of the right Rapid Alternating Movements of Hands. (e) Shap value of each level of the right Rapid Alternating Movements of Hands.

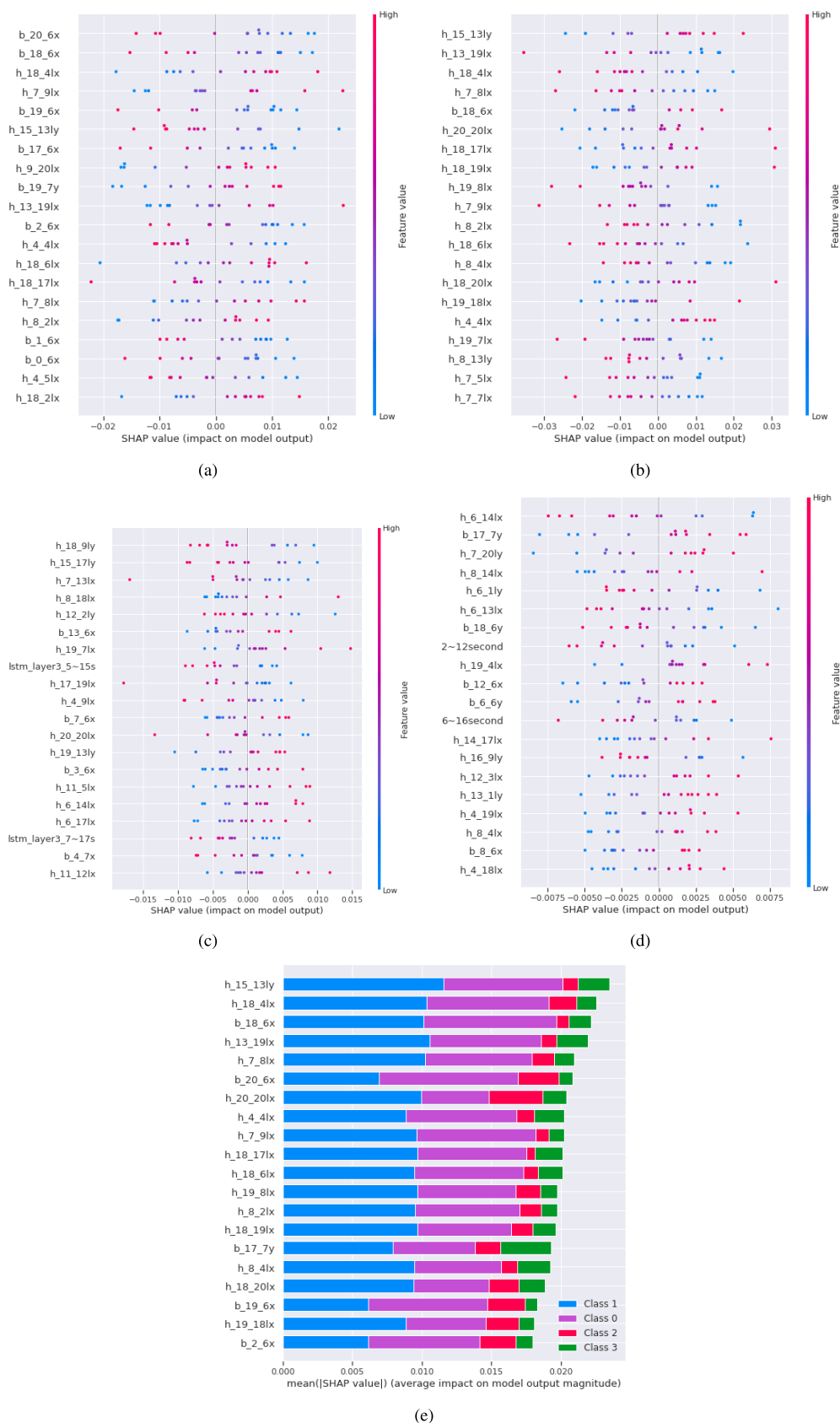


FIGURE 15. (a) Level 0 shap value of the left Rapid Alternating Movements of Hands; (b) Level 1 shap value of the left Rapid Alternating Movements of Hands; (c) Level 2 shap value of the left Rapid Alternating Movements of Hands; (d) Level 3 shap value of the left Rapid Alternating Movements of Hands.(e) Shap value of each level of the left Rapid Alternating Movements of Hands.

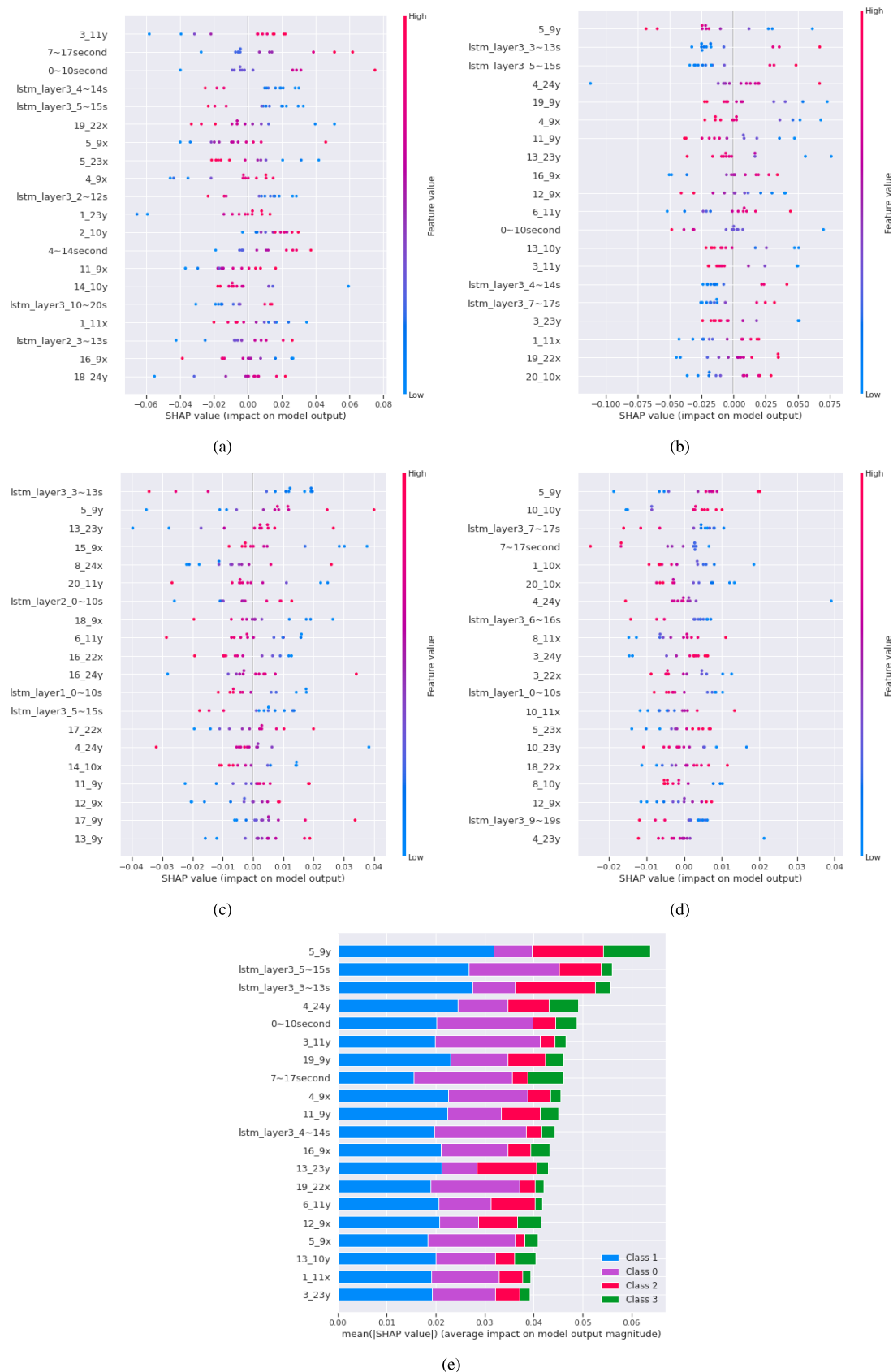


FIGURE 16. (a) Level 0 shap value of the right Leg Agility; (b) Level 1 shap value of the right Leg Agility; (c) Level 2 shap value of the right Leg Agility; (d) Level 3 shap value of the right Leg Agility. (e) Shap value of each level of the right Leg Agility.

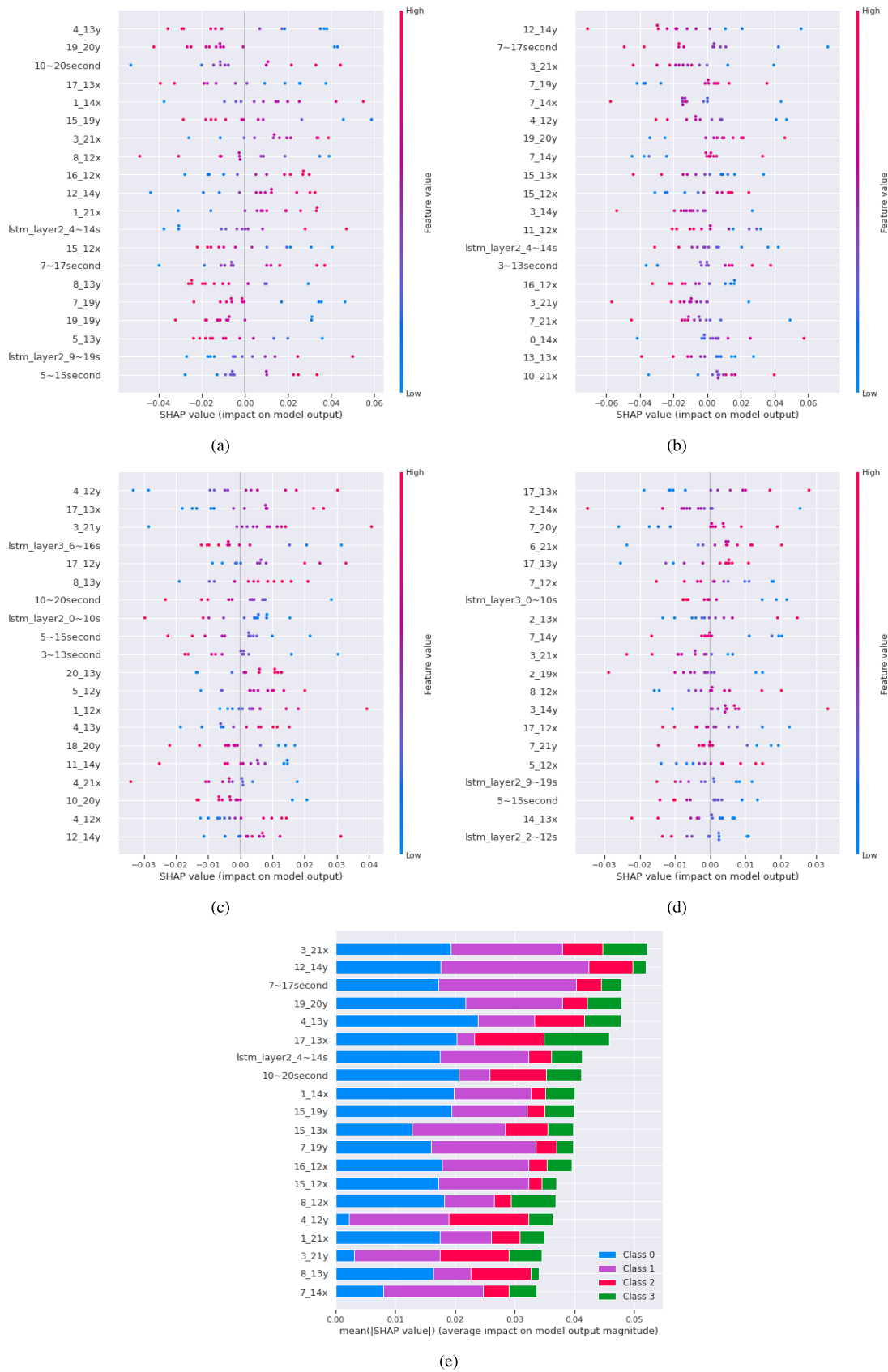


FIGURE 17. (a) Level 0 shap value of the left Leg Agility; (b) Level 1 shap value of the left Leg Agility; (c) Level 2 shap value of the left Leg Agility; (d) Level 3 shap value of the left Leg Agility. (e) Shap value of each level of the left Leg Agility.

REFERENCES

- [1] J. M. Beitz, "Parkinson's disease: A review," *Frontiers Bioscience-Scholar*, vol. 6, pp. 65–74, Jan. 2014.
- [2] C. Robertson and K. Flowers, "Motor set in Parkinson's disease," *J. Neurol., Neurosurgery Psychiatry*, vol. 53, pp. 583–592, Jul. 1990.
- [3] S.-T. Vista. (2012). *New Breakthrough in Parkinson's Disease Diagnosis*. [Online]. Available: <https://scitechvista.nat.gov.tw/Article/C000003/detail?ID=d456ac01-88af-4a4b-911c-abd8786556fa>
- [4] M. Heijmans, J. G. V. Habets, C. Herff, J. Aarts, A. Stevens, M. L. Kuijf, and P. L. Kubben, "Monitoring Parkinson's disease symptoms during daily life: A feasibility study," *NPJ Parkinson's Disease*, vol. 5, no. 1, p. 21, Sep. 2019.
- [5] InsightXplorer. (2022). *Insightxplorer Biweekly Report 2022*. [Online]. Available: https://www.ixresearch.com/wp-content/uploads/2022/02/InsightXplorer-Biweekly-Report_20220215.pdf
- [6] C.-M. Chang, Y.-L. Huang, J.-C. Chen, and C.-C. Lee, "Improving automatic tremor and movement motor disorder severity assessment for Parkinson's disease with deep joint training," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2019, pp. 3408–3411.
- [7] H. Li, X. Shao, C. Zhang, and X. Qian, "Automated assessment of parkinsonian finger-tapping tests through a vision-based fine-grained classification model," *Neurocomputing*, vol. 441, pp. 260–271, Jun. 2021.
- [8] M. H. Li, T. A. Mestre, S. H. Fox, and B. Taati, "Vision-based assessment of parkinsonism and levodopa-induced dyskinesia with pose estimation," *J. NeuroEng. Rehabil.*, vol. 15, no. 1, pp. 1–13, Nov. 2018.
- [9] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, "Realtime multi-person 2D pose estimation using part affinity fields," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1302–1310.
- [10] C. G. Goetz, B. C. Tilley, S. R. Shaftman, G. T. Stebbins, S. Fahn, P. Martinez-Martin, W. Poewe, C. Sampaio, M. B. Stern, and R. Dodel, "Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): Scale presentation and clinimetric testing results," *Movement Disorders, Off. J. Movement Disorder Soc.*, vol. 23, pp. 2129–2170, Nov. 2008.
- [11] I. Parkinson and M. D. Society. (2008). *MDS-UPDRS*. [Online]. Available: <https://www.movementdisorders.org/MDS/MDS-Rating-Scales/MDS-Unified-Parkinsons-Disease-Rating-Scale-MDS-UPDRS.htm>
- [12] T. M. D. Society. (2020). *MDS-UPDRS*. [Online]. Available: <http://www.tmds.org.tw/knowledge/info.asp?/1.html>
- [13] O. Martinez-Manzanera, E. Roosma, M. Beudel, R. W. K. Borgemeester, T. van Laar, and N. M. Maurits, "A method for automatic and objective scoring of bradykinesia using orientation sensors and classification algorithms," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 5, pp. 1016–1024, May 2016.
- [14] J. C. Pérez-Ibarra, A. A. G. Siqueira, and H. I. Krebs, "Identification of gait events in healthy and Parkinson's disease subjects using inertial sensors: A supervised learning approach," *IEEE Sensors J.*, vol. 20, no. 24, pp. 14984–14993, Dec. 2020.
- [15] H. Jeon, W. Lee, H. Park, H. Lee, S. Kim, H. Kim, B. Jeon, and K. Park, "Automatic classification of tremor severity in Parkinson's disease using a wearable device," *Sensors*, vol. 17, no. 9, p. 2067, Sep. 2017.
- [16] E. Balaji, D. Brindha, V. K. Elumalai, and R. Vikrama, "Automatic and non-invasive Parkinson's disease diagnosis and severity rating using LSTM network," *Appl. Soft Comput.*, vol. 108, Sep. 2021, Art. no. 107463.
- [17] S.-E. Wei, V. Ramakrishna, T. Kanade, and Y. Sheikh, "Convolutional pose machines," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4724–4732.
- [18] P. Schwab and W. Karlen, "PhoneMD: Learning to diagnose Parkinson's disease from smartphone data," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, Jul. 2019, pp. 1118–1125.
- [19] S.-T. Tang, C.-H. Tai, C.-Y. Yang, and J.-H. Lin, "Feasibility of smartphone-based gait assessment for parkinson's disease," *J. Med. Biol. Eng.*, vol. 40, pp. 582–591, Jul. 2020.
- [20] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jan. 2017, pp. 1–12.
- [21] OpenCV. *OpenCV Videowriter Class Reference*. [Online]. Available: https://docs.opencv.org/3.4/dd/d9e/classcv_1_1VideoWriter.html
- [22] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.



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